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COMMUNICATION NODE, COMMUNICATION SYSTEM AND OPERATING METHOD THEREOF

Abstract

Provided are a method of operating a communication system using physical layer division, the method comprises extracting an initial random access signal from an uplink signal, in a first physical layer of a first communication node of the communication system reducing a data amount of the extracted initial random access signal in the first physical layer and calculating a time synchronization error using the initial random access signal with a reduced amount of data, in a second physical layer of a second communication node of the communication system.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/776,844, filed May 13, 2022, which is a National Stage of International Application No. PCT/KR2020/016476, filed Nov. 20, 2020, claiming priority to Korean Patent Application No. 10-2019-0152580, filed Nov. 25, 2019 and Korean Patent Application No. 10-2020-0156047, filed Nov. 19, 2020.

TECHNICAL FIELD

[0002] The present invention relates to a communication node, a communication system, and a method of operating the same, and more particularly, to a communication node capable of receiving an initial random access signal with a reduced amount of data extracted from an uplink signal in a first physical layer from another communication node and calculating a time synchronization error using the initial random access signal with the reduced amount of data in a second physical layer, a communication system, and a method of operating the same.

BACKGROUND ART

[0003] In the existing mobile communication system, in order to provide a communication service to a base station shadow area such as a building or subway, a distributed antenna structure is constructed with a remote wireless device, and the remote wireless device and a digital device of a base station are connected to each other using a matching unit.

[0004] In particular, when implementing a distributed antenna system in a 5G mobile communication system, there is a limit to installing an additional dedicated cable for 5G communication, so the cable installed for 3G/4G communication is often shared. However, there is a problem that the cable installed for 3G/4G communication has insufficient capacity to support the maximum transmission speed defined in the 5G standard specification.

DESCRIPTION OF EMBODIMENTS

Technical Problem

[0005] The present invention provides a communication node capable of receiving an initial random access signal with a reduced amount of data extracted from an uplink signal in a first physical layer from another communication node and calculating a time synchronization error using an initial random access signal with a reduced amount of data in a second physical layer, a communication system, and a method of operating the same.

Solution to Problem

[0006] According to an aspect of an embodiment, a method of operating a communication system using physical layer division may comprise extracting an initial random access signal from an uplink signal, in a first physical layer of a first communication node of the communication system, reducing a data amount of the extracted initial random access signal in the first physical layer and calculating a time synchronization error using the initial random access signal with a reduced

amount of data, in a second physical layer of a second communication node of the communication system.

[0007] According to an aspect of an embodiment, the first physical layer may be a lower physical layer according to the physical layer division, and the second physical layer may be an upper physical layer according to the physical layer division.

[0008] According to an aspect of an embodiment, the initial random access signal may be a physical random access channel (PRACH) signal.

[0009] According to an aspect of an embodiment, the first communication node may be a remote unit, and the second communication node may be a distributed unit.

[0010] According to an aspect of an embodiment, the extracting of the initial random access signal from the uplink signal may comprise shifting a frequency band of the uplink signal to a baseband and extracting the initial random access signal by low-pass filtering the frequency band-shifted uplink signal.

[0011] According to an aspect of an embodiment, the reducing of the data amount of the extracted initial random access signal may comprise downsampling the extracted initial random access signal to reduce the data amount of the initial random access signal.

[0012] According to an aspect of an embodiment, the method may further comprise transmitting a combined signal generated by combining the uplink signal through data channel processing of the first physical layer in the first communication node and the initial random access signal with the reduced amount of data to the second communication node, and the calculating of the time synchronization error may comprise separating the initial random access signal with the reduced amount of data from the combined signal and calculating a time synchronization error using the separated initial random access signal with the reduced amount of data.

[0013] According to an aspect of an embodiment, the calculating of the time synchronization error may comprise calculating a first time offset using the initial random access signal with the reduced amount of data, restoring an initial random access signal with an original amount of data by using the initial random access signal with the reduced amount of data and calculating a second time offset using the first time offset and the restored initial random access signal.

[0014] According to an aspect of an embodiment, the calculating of the first time offset may comprise calculating a correlation value between the initial random access signal with the reduced amount of data and a plurality of random access sequences, searching for a random access sequence having a maximum correlation value from among the plurality of random access sequences and calculating the first time offset using the searched random access sequence having the maximum correlation value.

[0015] According to an aspect of an embodiment, the restoring of the initial random access signal with the original amount of data may comprise upsampling the initial random access signal with the reduced amount of data and interpolating the upsampled initial random access signal.

[0016] According to an aspect of an embodiment, the interpolating of the upsampled initial random access signal may comprise low-pass filtering the upsampled initial random access signal.

[0017] According to an aspect of an embodiment, the calculating of the second time offset using the first time offset and the restored initial random access signal may comprise calculating the second time offset using the first time offset, the restored initial random access signal, and the random access sequence having the maximum correlation value.

[0018] According to an aspect of an embodiment, the calculating of the second time offset using the first time offset, the restored initial random access signal, and the random access sequence having the maximum correlation value may comprise calculating the second time offset based on a point at which a correlation value of the restored initial random access signal and the random access sequence having the maximum correlation value becomes maximum within a reference range from the first time offset.

[0019] According to an aspect of an embodiment, a communication node may include a memory

and a processor, wherein the communication node, in a first physical layer of another communication node, may receive an initial random access signal with a reduced amount of data extracted from an uplink signal from the other communication node, and calculate a time synchronization error using the initial random access signal with the reduced amount of data in a second physical layer of the communication node.

[0020] According to an aspect of an embodiment, a communication system using physical layer division may comprise a first communication node configured to extract an initial random access signal from an uplink signal and reducing a data amount of the extracted initial random access signal in a first physical layer and a second communication node configured to calculate a time synchronization error using the initial random access signal with a reduced amount of data in a second physical layer.

Advantageous Effects of Disclosure

[0021] A method and a device according to an embodiment of the present invention have an advantage in that the capacity of data (e.g., a physical random access channel (PRACH) signal) required to be transmitted between communication nodes may be reduced in a process of calculating a time synchronization error, and at the same time, because the time synchronization error is estimated in an upper physical layer, information for calculating the time synchronization error (e.g., information about random access sequences, etc.) does not need to be shared with a lower physical layer.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0022] A brief description of each drawing is provided to more fully understand drawings recited in the detailed description of the present invention.

[0023] FIG. 1 is a conceptual diagram of a communication system according to an embodiment of the present invention.

[0024] FIG. 2 is a block diagram of a distributed unit shown in FIG. 1.

[0025] FIG. 3 is a block diagram of a first physical layer processor and a remote unit of the distributed unit shown in FIG. 1.

[0026] FIG. 4 is an embodiment of an initial random access signal processor included in a second physical layer processor shown in FIG. 3.

[0027] FIG. 5 is an embodiment of an initial random access signal included in an uplink signal.

[0028] FIG. 6 is an embodiment of a time synchronization error detector included in the first physical layer processor shown in FIG. 3.

[0029] FIG. 7 is a flowchart illustrating a method of operating a communication system according to an embodiment of the present invention.

MODE OF DISCLOSURE

[0030] The inventive concept may be variously modified and have various embodiments, so that specific embodiments will be illustrated in the drawings and described in the detailed description. However, this does not limit the inventive concept to specific embodiments, and it should be understood that the inventive concept covers all the modifications, equivalents and replacements included within the idea and technical scope of the inventive concept.

[0031] In describing the inventive concept, in the following description, a detailed explanation of known related technologies may be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure. In addition, numeral figures (e.g., 1, 2, and the like) used during describing the specification are just identification symbols for distinguishing one element from another element.

[0032] Further, in the specification, if it is described that one component is “connected” or

“accesses” the other component, it is understood that the one component may be directly connected to or may directly access the other component but unless explicitly described to the contrary, another component may be “connected” or “access” between the components.

[0033] In addition, terms including “unit,” “er,” “or,” “module,” and the like disclosed in the specification mean a unit that processes at least one function or operation and this may be implemented by hardware or software such as a processor, a micro processor, a micro controller, a central processing unit (CPU), a graphics processing unit (GPU), an accelerated Processing unit (APU), a digital signal processor (DSP), an application specific integrated circuit (ASIC), and a field programmable gate array (FPGA) or a combination of hardware and software. Furthermore, the terms may be implemented in a form coupled to a memory that stores data necessary for processing at least one function or operation.

[0034] Moreover, it is intended to clarify that components in the specification are distinguished in terms of primary functions of the components. That is, two or more components to be described below may be provided to be combined to one component or one component may be provided to be divided into two or more components for each more subdivided function. In addition, each of the respective components to be described below may additionally perform some or all functions among functions which other components take charge of in addition to a primary function which each component takes charge of and some functions among the primary functions which the respective components take charge of are exclusively charged by other components to be performed, of course.

[0035] FIG. 1 is a conceptual diagram of a communication system according to an embodiment of the present invention.

[0036] Referring to FIG. 1, a communication system **10** according to an embodiment of the present invention may include a plurality of remote units (RUs) **100**, a distributed unit (DU) **200**, a centralized unit (CU) **300**, and a core network.

[0037] The plurality of remote units **100** may be connected to the distributed unit **200** and distributed to transmit and receive signals transmitted/received from a base station at locations distributed in various shaded areas or dense areas.

[0038] The communication system **10** according to an embodiment of the present invention may use physical layer division between the plurality of remote units **100** and the distributed unit **200**. The plurality of remote units **100** may process a lower physical layer (Low-PHY), and the distributed unit **200** may process an upper physical layer (High-PHY).

[0039] According to an embodiment, the plurality of remote units **100** and the distributed unit **200** may be connected to each other through various communication media such as optical cables, high-speed cables, or microwave cables.

[0040] According to an embodiment, the plurality of remote units **100** and the distributed unit **200** may conform to various fronthaul communication standards. For example, an Ethernet-based Common Public Radio Interface (eCPRI), which is a 5G fronthaul interface standard, may be used between the plurality of remote units **100** and the distributed unit **200**.

[0041] The base station may be divided into the distributed unit **200** and the centralized unit **300**, wherein the distributed unit **200** and the centralized unit **300** may separate functions in various forms according to various function split options (e.g., 3GPP TR38.801 standard, etc.).

[0042] According to an embodiment, the distributed unit **200** and the centralized unit **300** may divide and process layers into RRC, PDCP, High-RLC, Low-RLC, High-MAC, Low-MAC, and High-PHY.

[0043] FIG. 1 illustrates a structure in which the base station is divided into the distributed unit **200** and the centralized unit **300**, but is not limited thereto, and various modifications are possible.

[0044] The centralized unit **300** may directly work with a core network, and may interface between the core network and the base station.

[0045] Detailed structures and operations of the remote unit **100** and the distributed unit **200** will

be described later with reference to FIGS. 2 to 7.

[0046] FIG. 2 is a block diagram of the distributed unit shown in FIG. 1.

[0047] Referring to FIGS. 1 and 2, the distributed unit **200** may include a power supply **210**, a controller **220**, a transfer device **230**, and a first physical layer processor **240**.

[0048] The power supply **210** may supply power to components (e.g., **220** to **240**) in the distributed unit **200**.

[0049] The controller **220** may control all operations of the components (e.g., **210**, **230**, and **240**) in the distributed unit **200**.

[0050] The transfer device **230** may interface communication between the distributed unit **200** and the centralized unit **300**, and may perform signal processing necessary for signal transmission/reception with the centralized unit **300**.

[0051] The first physical layer processor **240** may process functions supported by the first physical layer (e.g., the upper physical layer (High-PHY)).

[0052] Detailed structure and operation of the first physical layer processor **240** will be described later with reference to FIGS. 3 and 5.

[0053] FIG. 3 is a block diagram of a first physical layer processor and a remote unit of the distributed unit shown in FIG. 1. FIG. 4 is an embodiment of an initial random access signal processor included in a second physical layer processor shown in FIG. 3. FIG. 5 is an embodiment of an initial random access signal included in an uplink signal. FIG. 6 is an embodiment of a time synchronization error detector included in the first physical layer processor shown in FIG. 3.

[0054] Referring to FIGS. 1 to 3, the remote unit **100** may include an RF front-end **110**, an analog-to-digital converter (ADC) **120**, and a second physical layer processor **130**.

[0055] The remote unit **100** may convert an uplink signal of an RF band received through an antenna into a baseband signal through the RF front-end **110**.

[0056] The ADC **120** may convert an analog baseband signal converted by the RF front-end **110** into a digital signal.

[0057] The ADC **120** may transmit the converted digital signal to the second physical layer processor **130**.

[0058] The second physical layer processor **130** may include a data channel processor **132**, an initial random access signal processor **134**, and a multiplexer (MUX) **136**.

[0059] The second physical layer processor **130** may process a portion of a physical layer (e.g., the lower physical layer (low-PHY)) divided in the communication system **10** using physical layer division.

[0060] The data channel processor **132** may process a data channel of a digitally converted uplink signal. According to an embodiment, the data channel processor **132** may perform processing such as cyclic prefix (CP) removal, discrete Fourier transform (DFT), and data compression on the uplink signal.

[0061] The initial random access signal processor **134** may extract an initial random access signal from the digitally converted uplink signal and reduce a data amount of the extracted initial random access signal.

[0062] According to an embodiment, the initial random access signal may be a Physical Random Access Channel (PRACH) signal.

[0063] Referring to FIG. 4 together, the initial random access signal processor **134** may include a mixer **134-1**, a low pass filter **134-2**, and a down sampler **134-3**.

[0064] The mixer **134-1** may receive an uplink signal $x(n)$ that is digitally converted and transmitted by the ADC **120**, and may shift a frequency band of the received uplink signal $x(n)$ to the baseband.

[0065] The input signal $x(n)$ and an output signal $x_b(n)$ input/output by the mixer **134-1** may be expressed by the relationship of Equation 1 below.

[00001] $x_B(n) = x(n)e^{-j2\pi \frac{f_{RA}n}{N}}$ (Equation1) [0066] (where N is a DFT size when using orthogonal frequency division multiplexing (OFDM), and n is a time index of a sample)

[0067] Referring to FIG. 5 together, in an uplink signal, a frequency band in which an initial random access signal (e.g., a PRACH signal) is located is shown, a center frequency is $f_{sub.RA}$, and a bandwidth from the center frequency to both ends of the frequency band is denoted by W.

[0068] The mixer **134-1** may frequency shift the uplink signal (e.g., the PRACH signal) so that the center frequency of the initial random access signal becomes zero.

[0069] The low-pass filter **134-2** may perform low-pass filtering so that only the frequency band in which the initial random access signal (e.g., the PRACH signal) exists in the uplink signal in which the frequency band is shifted to a baseband by the mixer **134-1** passes through.

[0070] The low-pass filter **134-2** may extract only the initial random access signal (e.g., the PRACH signal) from the uplink signal through the low-pass filtering and output the initial random access signal as an output signal $y(n)$.

[0071] In this case, the output signal $y(n)$ may be expressed as in Equation 2 below.

[00002] $y(n) = \sum_{m=0}^{N_{LPF}-1} h(m)x_B(n-m)$ (Equation2) [0072] (where $h(n)$ is an impulse response

of the low-pass filter **134-2**, and $N_{sub.LPF}$ is the number of taps of the low-pass filter **134-2**)

[0073] According to an embodiment, a range of a pass band of the low pass filter **134-2** may be set according to a frequency range of the initial random access signal (e.g., the PRACH signal) frequency shifted to a baseband.

[0074] Returning to FIG. 4, the down-sampler **134-3** downsamples the output signal $y(n)$ filtered by the low-pass filter **134-2** to output an initial random access signal $y_{sub.D}(m)$ with a reduced amount of data.

[0075] According to an embodiment, the down-sampler **134-3** may downsample the output signal $y(n)$ filtered by the low-pass filter **134-2** by L times and output the output signal $y(n)$. In this case, the output signal may be expressed as in Equation 3 below.

[00003] $y_D(m) = y(mL + \alpha)$ (Equation3) [0076] (where m is an integer indicating a time index after downsampling, L is an integer satisfying $1 \leq L \leq B/W$, and α is an integer value satisfying $1 \leq \alpha \leq L-1$, which is a downsampling offset)

[0077] According to an embodiment, the down sampler **134-3** may be modified to reduce the amount of data in various ways (e.g., a compressor for compressing data, etc.).

[0078] Returning to FIG. 3, the multiplexer **136** may combine an uplink signal processed and output by the data channel processor **132** and an initial random access signal extracted by the initial random access signal processor **134** to reduce the amount of data to generate a combined signal, and may output the generated combined signal.

[0079] According to an embodiment, the multiplexer **136** may combine the uplink signal processed and output by the data channel processor **132** and the initial random access signal extracted by the initial random access signal processor **134** to reduce the amount of data into a single data stream.

[0080] The multiplexer **136** may transmit the combined signal to the first physical layer processor **240**.

[0081] The first physical layer processor **240** may include a demultiplexer (DEMUX) **250**, a data channel processor **260**, and a time synchronization error detector **270**.

[0082] The demultiplexer **250** may separate an initial random access signal with a reduced amount of data from the combined signal transmitted from the multiplexer **136** of the second physical layer processor **130**.

[0083] The demultiplexer **250** may transmit the uplink signal that has been processed and output by the data channel processor **132** of the second physical layer processor **130** to the data channel processor **260** of the first physical layer processor **240**, and the initial random access signal with a

reduced amount of data may be transmitted to the time synchronization error detector **270**.

[0084] The data channel processor **260** may receive the uplink signal that has been processed and output by the data channel processor **132** of the second physical layer processor **130** to perform data channel processing in the first physical layer.

[0085] According to an embodiment, the data channel processor **260** may perform various processes such as demodulation, demapping, and data decompression on the uplink signal.

[0086] The time synchronization error detector **270** may include a correlator **271**, a random access (RA) sequence detector **272**, a first time offset estimator **273**, a frequency offset compensator **274**, and a second time offset estimator **275**.

[0087] The correlator **271** may calculate a correlation value between a plurality of initial random access signals with a reduced amount of data and a plurality of random access sequences.

[0088] The random access sequence detector **272** may search for a random access sequence having a maximum correlation value from among the plurality of random access sequences, and may output information about the maximum correlation value and a random access sequence corresponding thereto (e.g., an index) according to a search result.

[0089] The first time offset estimator **273** may calculate a first time offset using a random access sequence having the maximum correlation value.

[0090] The frequency offset compensator **274** may estimate a frequency offset of a received initial random access signal and compensate the estimated frequency offset to output an initial random access signal in which the frequency offset is compensated.

[0091] The second time offset estimator **275** may calculate the first time offset output by the first time offset estimator **273**, a restored initial random access signal obtained by restoring the initial random access signal with a reduced amount of data, and a second time offset using the random access sequence having the maximum correlation value.

[0092] For description of the detailed structure and operation of the time synchronization error detector **270**, referring to FIG. 6, the correlator **271** may include a plurality of correlators **271-1** to **271-J**.

[0093] Each of the plurality of correlators **271-1** to **271-J** (where J is an integer of 2 or more) may calculate and output a correlation value between each of a plurality of random access sequences generated by a random access sequence generator **272-3** and the initial random access signal $y_{\text{sub.D}}(m)$ with a reduced amount of data.

[0094] The random access sequence detector **272** may include a plurality of peak detectors **272-11** to **272-1J**, a random access sequence index detector **272-2**, the random access sequence generator **272-3**, and a random access sequence selector **272-4**.

[0095] The plurality of peak detectors **272-11** to **272-1J** may detect and output maximum correlation values V_1 to V_J from correlation values output from the plurality of correlators **271-1** to **271-J**, respectively.

[0096] The plurality of peak detectors **272-11** to **272-1J** may detect and output time values $\tau_{\text{sub.1}}$ to $\tau_{\text{sub.J}}$ with respect to points corresponding to the maximum correlation values V_1 to V_J from the correlation values output from the plurality of correlators **271-1** to **271-J**, respectively.

[0097] The random access sequence index detector **272-2** may detect a largest correlation value V_{max} from among the maximum correlation values V_1 to V_J by using the maximum correlation values V_1 to V_J respectively output from the plurality of peak detectors **272-11** to **272-1J**, and output an index value $I_{\text{sub.RA}}$ of a random access sequence corresponding to the largest correlation value V_{max} .

[0098] The random access sequence generator **272-3** may generate and output a plurality of random access sequences that are candidates.

[0099] The random access sequence selector **272-4** may select and output a random access sequence corresponding to the index value $I_{\text{sub.RA}}$ of the random access sequence having the largest correlation value V_{max} from among the plurality of random access sequences generated by

the sequence generator **272-3**.

[0100] The first time offset estimator **273** may receive the time values $\tau_{\text{sub}.1}$ to $\tau_{\text{sub}.J}$ from the plurality of peak detectors **272-11** to **272-1J**, receive the index value $I_{\text{sub}.RA}$ from the random access sequence index detector **272-2**, and calculate the time value $\tau_{\text{sub}.c}$ corresponding to the index value $I_{\text{sub}.RA}$ from among the time values $\tau_{\text{sub}.1}$ to $\tau_{\text{sub}.J}$ as the first time offset.

[0101] The frequency offset compensator **274** may estimate and compensate the frequency offset using the first time offset $\tau_{\text{sub}.c}$ calculated by the first time offset estimator **273** and a repetition characteristic of an initial random access signal (e.g., the PRACH signal) in a time domain.

[0102] The second time offset estimator **275** may include a signal restorer **275-1**, a correlator **275-2**, and a peak detector **275-3**.

[0103] The signal restorer **275-1** may include an up-sampler **275-1A** and a low-pass filter **275-1B**.

[0104] The up-sampler **275-1A** may upsample an initial random access signal with a reduced amount of data.

[0105] According to an embodiment, the up-sampler **275-1A** may perform upsampling by L times corresponding to a multiple of reducing a data amount of the initial random access signal in a second physical layer.

[0106] The low-pass filter **275-1B** may perform interpolation processing on the upsampled initial random access signal by low-pass filtering the upsampled initial random access signal.

[0107] According to an embodiment, an interpolated signal in the time domain may be obtained by performing low-pass filtering on the upsampled initial random access signal after converting the upsampled initial random access signal to a frequency domain using N-point DFT before the low-pass filtering of the low-pass filter **275-1B**, and applying the N-point IDFT to the low-pass filtered signal.

[0108] According to an embodiment, an interpolation method of the upsampled initial random access signal may be variously modified, such as nearest neighbor interpolation, linear interpolation, quadratic interpolation, cubic interpolation, and cubic interpolation for each section.

[0109] An initial random access signal with a reduced amount of data through processing of the up-sampler **275-1A** and the low-pass filter **275-1B** may be restored to the initial random access signal with the original amount of data.

[0110] The correlator **275-2** and the peak detector **275-3** may calculate the second time offset based on the first time offset $\tau_{\text{sub}.c}$, a restored initial random access signal $z(n)$, and a random access sequence $RA_{\text{sub}.X}$ having a maximum correlation value.

[0111] The correlator **275-2** may calculate and output a correlation value of the restored initial random access signal $z(n)$ and the random access sequence $RA_{\text{sub}.X}$ having the maximum correlation value within a reference range from the first time offset $\tau_{\text{sub}.c}$.

[0112] According to an embodiment, the reference range may be defined as in Equation 4 below.

[00004] $c - T_L \leq \quad \leq c + T_R$ (Equation 4) [0113] (where the T_L is a reference range of time fluctuation in the left direction, and the T_R is a reference range of time fluctuation in the right direction)

[0114] The peak detector **275-3** may calculate the second time offset based on a point at which a correlation value of the correlator **275-2** becomes maximum within the reference range.

[0115] According to an embodiment, the first time offset may have a meaning as a rough time synchronization error because an initial random access signal with a reduced amount of data is used, and the second time offset may have a meaning as a precise time synchronization error because an initial random access signal with the original amount of data restored is used. According to an embodiment, a time synchronization error in the present specification may be the first time offset or the second time offset.

[0116] FIG. 7 is a flowchart illustrating a method of operating a communication system according to an embodiment of the present invention.

[0117] Referring to FIGS. 1 to 7, in operation S710, the communication system **10** may extract an initial random access signal from an uplink signal in a first physical layer (e.g., the lower physical layer (low-PHY)) of a first communication node (e.g., the remote unit **100**).

[0118] According to an embodiment, the communication system **10** may extract the initial random access signal by performing low-pass filtering after frequency-shifting the uplink signal.

[0119] According to an embodiment, the initial random access signal may be a PRACH signal.

[0120] In operation S720, the communication system **10** may reduce a data amount of the extracted initial random access signal in the first physical layer (e.g., the lower physical layer (low-PHY)).

[0121] According to an embodiment, the communication system **10** may reduce the amount of data by downsampling the extracted initial random access signal.

[0122] In operation S730, the communication system **10** may calculate a time synchronization error using the initial random access signal with a reduced amount of data in the second physical layer (e.g., the upper physical layer (high-PHY)) of a second communication node (e.g., the distributed unit **200**).

[0123] According to an embodiment, the communication system **10** may obtain a first time offset using the initial random access signal with a reduced amount of data, and may obtain a second time offset using the first time offset and an initial random access signal with the original amount of data restored.

[0124] According to an embodiment, the time synchronization error calculated by the communication system **10** may mean the first time offset or the second time offset. While the present disclosure has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the scope of the following claims.

Claims

1. A method of operating a communication system using physical layer division, the method comprising: extracting an initial random access signal from an uplink signal, in a first physical layer of a first communication node of the communication system; reducing a data amount of the extracted initial random access signal in the first physical layer; transmitting the initial random access signal with the reduced amount of data from the first communication node to a second communication node of the communication system; and calculating a time synchronization error between the first communication node and the second communication node using the initial random access signal with the reduced amount of data, in a second physical layer of the second communication node, wherein the extracting of the initial random access signal from the uplink signal comprises: shifting a frequency band of the uplink signal to a baseband; and extracting the initial random access signal by low-pass filtering the frequency band-shifted uplink signal.
2. The method of claim 1, wherein the first physical layer is a lower physical layer according to the physical layer division, and the second physical layer is an upper physical layer according to the physical layer division.
3. The method of claim 1, wherein the initial random access signal is a physical random access channel (PRACH) signal.
4. The method of claim 1, wherein the first communication node is a remote unit, and the second communication node is a distributed unit.
5. The method of claim 1, wherein the reducing of the data amount of the extracted initial random access signal comprises: downsampling the extracted initial random access signal to reduce the data amount of the initial random access signal.
6. The method of claim 1, further comprising: transmitting a combined signal generated by combining the uplink signal through data channel processing of the first physical layer in the first communication node and the initial random access signal with the reduced amount of data to the

second communication node, and the calculating of the time synchronization error comprises: separating, by the second communication node, the initial random access signal with the reduced amount of data from the combined signal; and calculating, by the second communication node, the time synchronization error using the separated initial random access signal with the reduced amount of data.

7. The method of claim 1, wherein the calculating of the time synchronization error comprises: calculating a first time offset using the initial random access signal with the reduced amount of data; restoring an initial random access signal with an original amount of data by using the initial random access signal with the reduced amount of data; and calculating a second time offset using the first time offset and the restored initial random access signal.

8. The method of claim 7, wherein the calculating of the first time offset comprises: calculating a correlation value between the initial random access signal with the reduced amount of data and a plurality of random access sequences; searching for a random access sequence having a maximum correlation value from among the plurality of random access sequences; and calculating the first time offset using the searched random access sequence having the maximum correlation value.

9. The method of claim 8, wherein the restoring of the initial random access signal with the original amount of data comprises: upsampling the initial random access signal with the reduced amount of data; and interpolating the upsampled initial random access signal.

10. The method of claim 9, wherein the interpolating of the upsampled initial random access signal comprises: low-pass filtering the upsampled initial random access signal.

11. The method of claim 10, wherein the calculating of the second time offset using the first time offset and the restored initial random access signal comprises: calculating the second time offset using the first time offset, the restored initial random access signal, and the random access sequence having the maximum correlation value.

12. The method of claim 11, wherein the calculating of the second time offset using the first time offset, the restored initial random access signal, and the random access sequence having the maximum correlation value comprises: calculating the second time offset based on a point at which a correlation value of the restored initial random access signal and the random access sequence having the maximum correlation value becomes maximum within a reference range from the first time offset.

14. A communication system using physical layer division, the communication system comprising: a first communication node configured to shift a frequency band of the uplink signal to a baseband and extract the initial random access signal by low-pass filtering the frequency band-shifted uplink signal and reduce a data amount of the extracted initial random access signal in a first physical layer, and transmit an initial random access signal with the reduced amount of data to a second communication node; and the second communication node configured to calculate a time synchronization error between the first communication node and the second communication node using the initial random access signal with the reduced amount of data in a second physical layer.
